Bhargav M

+91 7795036940 | bhargav.m23@iiits.in | linkedin.com/in/bhargav-ai
github.com/bhargavm-iiits | bhargavm-portfolio.vercel.app

Professional Summary

B.Tech AI and Data Science student graduating in 2027, building AI-enabled full-stack applications: RAG pipelines, LangGraph workflows, LLM guardrails, and automated validation in Python, FastAPI, and React. Focused on grounded, auditable, and secure AI systems.

Technical Skills

Languages and Frameworks: Python, SQL, JavaScript/TypeScript, React, C#, FastAPI, REST APIs, Streamlit, Pandas, NumPy
AI and LLM: LLMs and embeddings, retrieval-augmented generation (RAG), prompt engineering, Pydantic structured outputs, LangGraph, LangChain, ChromaDB, BM25, Gemini API, Anthropic SDK, LiveKit Agents (tool calling), PyTorch
AI Quality and Governance: Guardrails, citation validation, prompt-injection testing, PII redaction (Presidio), human-in-the-loop review, audit logging, pytest, regression testing, CI quality gates
Data, Cloud and DevOps: PostgreSQL, SQLite, SQLAlchemy, Git, GitHub Actions (CI), Docker, Docker Compose, Playwright, Linux, Vercel, Render, asyncio

Projects

FixProof | Verified AI Code Repair with Sandboxed Testing GitHub
Python, FastAPI, React, Docker, Gemini API, BM25, pytest, Playwright, Vercel

- Built a **verification system for AI-generated bug fixes**: a seven-stage pipeline (prepare, baseline, retrieve, propose, gate, verify, select) with **BM25 retrieval** and **Gemini** patch proposals, a **FastAPI** backend, and a **React** frontend on Vercel.
- Enforced a **nine-check ordered gate** (scope, no test-file edits, no new imports, no dangerous calls) and ran candidates in a **Docker sandbox** (no network, read-only image, capped CPU/memory/PIDs); a fix is accepted only if the program's own tests pass.
- Added **437 pytest tests**, Docker verifier tests, and **Playwright** E2E tests; built a **24-bug fixture set** with a dev/holdout split and configuration hashing.

ClaimBridge | AI Document Review with Grounded Generation and Guardrails GitHub
Python, SQLite, Pydantic, Anthropic SDK, Gemini, BM25, PyMuPDF, Streamlit, pytest

- Built an **AI-native document-review workflow**: **PDF extraction** with page-level provenance, deterministic reconciliation against synthetic claim JSON, and drafts blocked when evidence was missing or conflicting.
- Implemented **grounded generation** with **BM25 retrieval** (payer/date filters) and **Pydantic structured outputs** through a switchable **Anthropic/Gemini** LLM layer with template fallback; enforced **citation-ID validation**, one repair attempt, and escalation.
- Added **guardrails** and **human-in-the-loop** review: a **prompt-injection** test fixture, an append-only **audit log** in SQLite, evidence-preserving corrections, and **pytest** coverage of reconciliation, citations, and the injection case.

Excel-to-App | Spreadsheet Processing and API Automation GitHub
Python, FastAPI, React, TypeScript, PostgreSQL, SQLAlchemy, Gemini, Docker, pytest

- Built a **Python spreadsheet-to-API pipeline** with **FastAPI REST endpoints** and a **React/TypeScript** web app, supporting **66 functions** on a six-table **PostgreSQL/SQLite** schema with immutable versions and execution audit trails.
- Added **Gemini schema suggestions** with invalid-reference checks and **human approval**; kept all numeric calculations deterministic.
- Implemented **automated validation** against LibreOffice: **481/481 matching comparisons** on a 37-case fixture benchmark; documented **73 pytest tests** and packaged the application with **Docker Compose**.

Experience

CYMAX - BTP Internship Jan 2026 – Present
Unity & XR Developer Intern Hyderabad, India

- Developed a Unity 360-degree video playback system for VR headsets, integrating spherical rendering, headset/controller input, and frame-accurate playback controls.
- Built a **Python backend** for secure VR streaming with **authentication, rate limiting, and encrypted delivery**, connected to a **React** and React Three Fiber playback client.
- Implemented **Argon2** password hashing, **JWT** sessions, and **AES** stream encryption; containerised the stack with **Docker Compose** and covered backend functionality with **pytest**.

IIIT Sri City - Summer Internship May 2026 – Jul 2026
XR & Simulation Intern Sri City, India

- Implemented offline AR navigation over a 7.29 km route, combining GPS, visual-inertial odometry, and compass data through a **Kalman filter**, with **A* routing** over a waypoint graph.
- Chained **GeoJSON** segments by endpoint proximity; identified a **1.17 km source-geometry gap** and marked it as interpolated to distinguish estimated data from surveyed geometry.
- Built and scaled 3D assets in Blender and animated them in Unreal Engine for an ISRO third launch-pad visualisation and simulation walkthrough.

Education

Indian Institute of Information Technology, Sri City Aug 2023 – May 2027
B.Tech in Artificial Intelligence and Data Science
BGS PU College, Sidlaghatta – Class XII: **92.3%** Apr 2020 – May 2022
BGS Public School – Class X: **97.6%**